

Product Identity & Traceability SaaS

Complete Development Flow & Module Architecture

Product Owner	Codenap – Professional IT Services
Product Type	Multi-Tenant SaaS Platform
Development Model	Proprietary Product Development
Document Version	V1.0

Internal product-development blueprint for building Codenap’s reusable SaaS platform. The platform is designed to serve multiple brands and manufacturers across industries such as Pashmina, textiles, timber, furniture, handicrafts, jewellery and other premium physical products.

1. Complete System Architecture

The product is divided into five layers: SaaS Management, Customer Tenants, Core Engine, Customer Experience, and external integrations.



2. Multi-Tenant Architecture

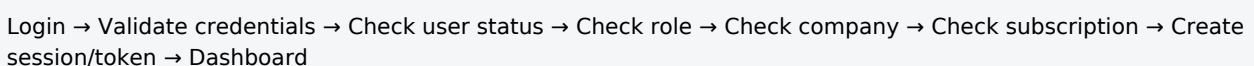
The SaaS must support many independent companies inside one platform while keeping company data isolated.



Every tenant-owned record should carry a tenant/company reference. The backend must derive tenant context from the authenticated user or API credential rather than trusting a company ID supplied by the browser.

3. Authentication & Role Management

SaaS User Authentication



Role-Based Access Control

Use reusable permissions instead of hard-coded role checks.

- product.view / product.create / product.edit / product.delete
- qr.view / qr.generate / qr.deactivate
- customer.view / customer.edit
- billing.view / settings.manage

Roles such as Company Admin, Product Manager, Customer Support, Service Manager and Viewer can be composed from these permissions.

4. Company Registration & Onboarding

SaaS Website → Select Plan → Create Account → Company Details → Industry → Brand Setup → Payment → Tenant Created → Admin Created → Subscription Activated → Onboarding → Dashboard

- Create company/tenant
- Create first company owner/admin
- Attach selected subscription plan
- Create default settings and branding
- Load industry-specific field configuration
- Activate trial or paid subscription

5. Subscription, Billing & Usage

Subscription

- Free Trial
- Monthly Subscription
- Annual Subscription
- Upgrade
- Downgrade
- Renew
- Cancel
- Suspend
- Reactivate

Billing Flow

Plan Selected → Payment Gateway → Verified Webhook → Subscription Updated → Invoice → Features Activated

Subscription activation should be based on backend verification/webhooks, not only a frontend payment-success message.

Usage Management

Operation Requested → Check Subscription → Check Usage Limit → Allowed? → Create Operation / Show Upgrade Prompt

- Products
- QR Codes
- QR Scans
- Storage
- API Requests
- Users

6. Codenap Super Admin Panel

Dashboard

- Total companies, active companies, trial companies, suspended companies
- Total products, QR codes, scans and customers
- Active subscriptions, MRR, ARR, new subscriptions and cancellations
- Company growth, revenue trends, product growth and scan trends

Company / Tenant Management

- Add/edit/view company
- Activate/suspend/archive
- Change plan
- View usage
- View subscription
- View activity

Plan & Subscription Management

- Create plan
- Configure limits
- Configure features
- Manage subscriptions
- View billing/payment status

Support

- Tickets
- Assignment
- Internal notes
- Attachments
- Status management

7. Company / Brand Admin Panel

Dashboard

- Products
- Active QR codes
- Customers
- Registrations
- Scans
- Successful/failed verification
- Suspicious scans
- Warranties
- Open service requests

Company Settings

- Logo
- Brand name
- Colors
- Website
- Contact information
- Support details

Team Management

- Invite users
- Assign roles
- Set permissions
- Deactivate users

8. Product Management

The product module is the central business entity of the SaaS.

Product Created → Product Identity → QR Assigned → Product Activated → Sold → Scanned → Registered → Warranty → Service

Add Product Flow

Basic Details → Identity → Industry Data → Passport Content → Warranty → Save → Product Created

- Basic: product name, SKU, category, collection, images, description
- Identity: product ID, serial number, batch, manufacturing date
- Industry data: dynamic fields based on selected industry
- Passport: story, certificate, care guide
- Warranty: assign warranty plan

Bulk Product Import

Download Template → Fill Excel/CSV → Upload → Validate → Preview → Confirm → Create Products

- Required-field validation
- Duplicate SKU/Product ID detection
- Date and numeric validation
- Error report
- Import summary

9. Product Identity & Serialization

Product IDs must be globally unique at platform level while allowing company-specific formats.

PAS-2026-000001
TIM-2026-000001

- Configurable prefix
- Year
- Sequence
- Optional category code
- SKU/serial/batch mapping

10. QR Code Engine

Single QR Flow

Product → Request QR → Check product → Check plan limit → Generate secure token → Create QR → Map QR → Activate

The QR should contain a secure verification URL/token rather than exposing sensitive product/company information.

Bulk QR Flow

Select Batch → Quantity → Check Limit → Generate IDs/Tokens → Background Job → QR Generation → Progress → ZIP/PDF

Large QR jobs should run asynchronously through a queue/worker rather than inside one long browser request.

QR Replacement

Old QR → Deactivate → Generate Replacement → Map New QR → Keep Product Identity

11. Product Authentication

Scan QR → Verification URL → Receive token → Find QR → Find Product → Check Company → Check QR/Product status → Risk rules → Create Scan Record → Return result

Result States

- Authentic: Product Verified
- Invalid: Unable to Verify Product
- Inactive: Verification Code Inactive
- Suspicious: Additional Verification Required

Authentication should be treated as verification against the platform's registered product identity; a QR alone should not be marketed as an absolute guarantee against counterfeiting.

12. Scan & Suspicious Activity Engine

Scan Record

- scanId
- companyId
- productId
- qrId
- timestamp
- approximate location where applicable
- device/browser information
- result
- risk information

Initial Risk Rules

- Excessive scans in a short period
- Repeated scans
- Unusual geographic pattern
- Impossible travel pattern
- Abnormal verification frequency

Start with rule-based risk scoring. Do not require AI in the first release.

Risk 0-30: Normal
31-60: Review
61-80: Suspicious
81-100: High Risk

A suspicious score should trigger review rather than automatically declaring a product counterfeit.

13. Digital Product Passport

The passport should be dynamically rendered from product data, company branding and a passport template rather than storing a separate page for every product.

Product Data + Brand Settings + Passport Template → Dynamic Customer Passport

- Product identity
- Product image
- Material
- Origin
- Manufacturing information

- Craftsmanship
- Product story
- Certifications
- Certificate
- Care guide
- Warranty
- Service history

14. Industry Configuration Engine

Do not build separate product databases for Pashmina, timber and jewellery. Use a generic Product entity plus configurable industry schemas.

Generic Product + Industry Schema → Industry-specific fields

Example: Pashmina

- Material
- Origin
- Craftsmanship
- Artisan
- Collection

Example: Timber

- Wood species
- Origin
- Grade
- Dimensions
- Volume
- Moisture
- Processing
- Certification

Example: Jewellery

- Material
- Purity
- Weight
- Stone
- Certificate
- Origin

Dynamic Field Engine

- Field name
- Field type
- Required flag
- Industry
- Validation
- Display order
- Customer visibility

15. Certificate Management

Product → Certificate Template → Certificate Instance → Certificate Number

- Create/edit templates
- Upload supporting certificate
- Generate certificate
- Assign to product
- Preview
- Download PDF
- Verification link/QR

Certificate generation should be tied to product data and should have clear rules for regeneration if underlying product information changes.

16. Care Guide

- Create reusable care guides
- Assign to product/category
- Display on customer passport

Examples include cleaning, washing, storage, maintenance, handling and do's/don'ts. Timber and Pashmina can use different care templates.

17. Customer Registration & Ownership

Registration Flow

Scan → Passport → Register Product → Enter Customer Details → Validate → Find Product → Check Registration → Create Customer → Create Ownership → Activate Warranty if configured → Confirmation

- Name
- Email
- Mobile
- Purchase date
- Order number
- Purchase channel
- Location
- Consent

Ownership Model

Use an ownership/purchase relationship instead of only storing one customer ID on a product. This keeps the system ready for future ownership transfer and resale.

Product → Ownership Record → Customer → Future Transfer → New Owner

18. Warranty Engine

Activation Rules

- Start on purchase date
- Start on registration
- Start on e-commerce order completion
- Start on installation

Product → Warranty Plan → Activation Event → Start Date → Duration → Expiry Date

Warranty Management

- Create plans
- Coverage
- Exclusions
- Duration
- Active/expired status
- Claims

19. Service Management

Customer Flow

My Product → Warranty/Service → Raise Request → Select Issue → Description → Photos/Documents → Submit

Admin Flow

New → Under Review → Approved → Assigned → In Progress → Resolved → Closed

Every status change should create a service activity/timeline entry.

Customer Timeline

✓ Request Submitted → ✓ Under Review → ✓ Approved → ● In Progress → ○ Resolved

20. Product Traceability

Traceability should use generic events rather than hard-coded manufacturing steps.

Traceability Event = eventType + product/batch + company + location + date + actor + quantity + documents + remarks

- Raw Material Received
- Manufactured
- Processed
- Quality Checked
- Warehouse
- Dispatched
- Distributor
- Sold
- Registered
- Serviced

Timber Example

Supplier/Forest → Raw Material → Processing → Kiln Drying → Quality Check → Warehouse → Exporter → Customer
→ Authentication → Digital Timber Passport

21. E-commerce Integration

E-commerce Order → SaaS API → Identify Product → Identify Customer → Create Ownership → Activate Warranty

The SaaS should integrate with Codenap-built or third-party e-commerce websites through APIs. The website should not duplicate the SaaS product identity, warranty or traceability logic.

22. API Architecture

Core API Groups

- Product API
- QR/authentication API
- Customer API
- Registration API
- Passport API
- Certificate API
- Warranty API
- Service API

API Authentication Flow

Company → API Key → API Request → Validate Key → Identify Tenant → Check Permission → Rate Limit → Process

Never rely only on a company ID sent by the client application to identify the tenant.

23. Webhooks & Integration Reliability

SaaS Event → Webhook Queue → Client Endpoint → Response → Retry on Failure → Log → Alert

- product.created
- product.registered
- qr.scanned
- warranty.activated
- service.created
- service.updated

Maintain integration logs with success, failed and retrying states.

24. White Label

Company → Brand Settings → Logo/Colors/Domain/Certificate/Passport → Customer Experience

- Custom logo
- Brand colors
- Custom verification page
- Custom certificate
- Custom domain
- Optional removal of SaaS branding

Example: verify.brand.com can use the same backend while presenting the customer's branding.

25. Notification Engine

Use a centralized notification service instead of calling email/SMS/WhatsApp functions directly from every module.

Module Event → Notification Service → Channel Selection → Email/SMS/WhatsApp

- product.registered
- warranty.expiring
- service.created
- subscription.failed

Third-party communication charges are separate from the SaaS software itself.

26. Audit Logs & Security

Audit Log

User → Action → Entity → Previous State → New State → Timestamp → IP/metadata where appropriate

- Login
- Product creation/update
- QR generation/deactivation
- Customer registration
- Certificate generation
- Warranty update
- Service update
- API activity

Security

- Tenant data isolation
- Secure authentication
- Role-based access
- API authentication
- Password hashing
- HTTPS
- Rate limiting
- Audit logs
- Database backups
- Secure QR tokens

27. Storage & File Management

Store large files such as product images, certificates, invoices and service photos in object storage. Keep metadata and secure references in the application database.

Application → Object Storage → Secure File Reference

28. Search, Reports & Analytics

Search

- Product ID
- SKU
- Serial number
- Batch
- QR ID
- Customer
- Order number

Search/filtering must be server-side so the admin panel can handle large product volumes.

Reporting Engine

Filter → Query → Dataset → Table/Chart → Export

- Product reports
- QR reports
- Authentication reports
- Customer reports

- Warranty reports
- Service reports
- Traceability reports

Large Export

Export Requested → Background Job → Generate CSV/XLSX → Store → Notify → Download

29. Database Core Entities

User / Role / Permission
 Company / Industry / Subscription / Plan / Usage / Invoice / Payment
 Product / Category / Collection / ProductField / ProductFieldValue
 QR / QRScan / AuthenticationEvent
 Certificate / CertificateTemplate
 CareGuide / CareGuideTemplate
 Customer / ProductRegistration / Ownership
 WarrantyPlan / Warranty / WarrantyClaim
 ServiceRequest / ServiceActivity
 TraceabilityEvent
 Integration / APIKey / Webhook / WebhookLog
 Notification / AuditLog

30. Complete Product Creation Flow

Company Admin
 ↓
 Add Product
 ↓
 Select Industry
 ↓
 Basic Details
 ↓
 Industry Fields
 ↓
 Images / Documents
 ↓
 Category / Collection
 ↓
 Generate Product ID
 ↓
 Save Product
 ↓
 Generate QR
 ↓
 Assign Passport / Certificate / Care Guide / Warranty
 ↓
 Activate Product

31. Complete Customer Flow

Customer Purchases Product
 ↓
 QR Scan
 ↓
 Authentication API
 ↓
 Product Found?
 YES ↓ NO ↓
 Risk Check Invalid
 ↓
 Verification

↓
Digital Passport
↓
Certificate / Care Guide
↓
Register Product
↓
Customer + Ownership
↓
Warranty
↓
Service if Required

32. Complete Timber Flow

Supplier / Forest
↓
Raw Material Batch
↓
Species / Origin
↓
Receiving
↓
Processing
↓
Quality Check / Grade / Moisture
↓
Certification
↓
Finished Product
↓
Product ID + QR/DataMatrix
↓
Warehouse / Dispatch
↓
Customer
↓
Authentication + Digital Timber Passport

33. Complete Pashmina Flow

Material / Production
↓
Batch
↓
Craftsmanship / Origin
↓
Product
↓
Product ID
↓
QR
↓
E-commerce Sale
↓
Customer
↓
Authentication
↓
Digital Passport + Certificate
↓
Registration

Codenap
↓
Product Identity & Traceability SaaS
↓
Pashmina / Timber / Furniture / Handicrafts / Jewellery / Manufacturing
↓
Subscription + Implementation + Integration + Custom Services

39. Master Development Roadmap

Foundation
→ Multi-tenancy
→ Roles & Permissions
→ Subscription
→ Product Engine
→ Industry Engine
→ QR Engine
→ Authentication
→ Digital Passport
→ Certificate
→ Customer
→ Warranty
→ Service
→ Traceability
→ Analytics
→ API
→ E-commerce
→ Enterprise
→ Security & Optimization
→ Production Launch

40. Core Product Principle

The entire SaaS should be designed around the relationship: Company → Product → Digital Identity → Customer. Every other module should attach to this core relationship.

COMPANY
|
└─ PRODUCT
└─ Product ID
└─ QR
└─ Passport
└─ Certificate
└─ Traceability
└─ Customer
└─ Warranty
└─ Service

If this core architecture is designed correctly, Codenap can add new industries without rebuilding the platform.